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ACTUAL ORDER OF CONVERGENCE OF RUNGE-KUTTA METHODS ON DIFFERENTIAL EQUATIONS WITH DISCONTINUITIES*

BRUCE A. CHARTRES† AND ROBERT S. STEPLEMAN‡

Abstract. This paper examines the order of convergence of Runge-Kutta methods for the ordinary differential equation $y'(t) = f(t, y(t))$ in Banach space, where the function f is badly behaved off the solution trajectory, and the solution y has derivatives only of low order. Under the assumption that f is everywhere Lipschitz continuous in its second argument, we get orders of convergence for the case when f has no derivatives at all, and for when it has only directional derivatives along the solution trajectory $y(t)$.

Utilization of the authors' general theory of convergence, stability and consistency enables us to obtain results which apply to actual computations in the presence of rounding error.

Some numerical experiments which support the conjecture that these convergence results are the best possible are briefly described.

1. Introduction. Let B be any real Banach space, and Y the space of continuous functions from $[0, 1]$ to B with uniform norm induced by the norm in B (i.e., $\|y\| = \sup \{\|y(t)\| : t \in [0, 1]\}$), and let $f: [0, 1] \times B \rightarrow B$. Then

$$(1.1) \quad y'(t) = f(t, y(t)), \quad y(0) = a,$$

is the ordinary differential equation (ODE) we wish to solve numerically. (The derivative is the classical one, defined as the limit, in the norm on B , of a difference quotient.) We shall not always require that f be a continuous function of t , nor that y be everywhere differentiable, hence we accept y as a solution of (1.1) if it satisfies

$$(1.2) \quad y(t) - a = \int_0^t f(s, y(s)) ds, \quad t \in [0, 1],$$

where we are using the Riemann integral for Banach-space-valued functions as defined by Graves [9]. Discontinuities can occur in the differential equations of initial value problems for a number of reasons. The closing or opening of a switch can lead to a completely new set of differential equations, and if the position of this switch is a function of one or more of the dependent variables, these discontinuities cannot be anticipated by any simple procedure. Saturation and backlash can also lead to discontinuities. We also require f and a to be such that a solution to (1.1) exists. We make no attempt to derive conditions for the existence and uniqueness of a solution to (1.2), being concerned only with the numerical analyst's problem of determining the conditions under which a given numerical method will converge, and its order of convergence. Hence we assume that at least one solution exists, and we let y denote any such solution. However, it is clear that in establishing convergence of a numerical method to y we are also incidentally proving the uniqueness of the solution, for a numerical method cannot converge to two distinct solutions. It should also be noted that since B is

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any real Banach space, our results will apply in particular to finite systems of equations.

We consider Runge–Kutta (R–K) methods as defined by the equations

$$(1.3) \quad \begin{aligned} y_{h,0} &= a, \\ y_{h,n+1} &= y_{h,n} + h \sum_{i=0}^m w_i k_{in}(y_{hn}) + \tau_{h,n+1}, \quad n = 0, \dots, N-1, \end{aligned}$$

where, for any $v \in B$,

$$(1.4) \quad k_{in}(v) = f\left(t_{h,n+\alpha_i}, v + h \sum_{j=0}^m \beta_{ij} k_{jn}(v)\right), \quad i = 0, \dots, m,$$

$hN = 1$, $t_{hn} = nh$, $y_{hn} = y(t_{hn})$ and the α_i , w_i and β_{ij} are real numbers with $0 \leq \alpha_i \leq 1$. Thus we are considering the full class of implicit R–K methods as defined by Butcher [1], including the classical explicit methods in which $\beta_{ij} = 0$ whenever $j \geq i$.

The τ_{hn} of (1.3) are the local truncation errors; their values are such that, if (1.4) is solved exactly for the k_{in} , then (1.3) is exactly satisfied by the solution y of (1.1). In any actual computation we do not obtain the y_{hn} but the “numerical solution” $\hat{y}_{hn}$ which satisfies

$$(1.5) \quad \begin{aligned} \hat{y}_{h0} &= a + e_{h0}, \\ \hat{y}_{h,n+1} &= \hat{y}_{hn} + h \sum_{i=0}^m w_i k_{in}(\hat{y}_{hn}) + e_{h,n+1}, \quad n = 0, \dots, N-1. \end{aligned}$$

The e_{hn} are “perturbations” that arise from rounding errors, errors of approximating f by a more easily computable function, and the error due to truncating an iterative method (if one is required) for solving (1.4) for k_{in} . Note that the functions k_{in} in (1.5) are the exact solutions of (1.4); they are not the calculated approximations to them (nor are they in general the $y'(t_{h,n+\alpha_i})$). All errors made in computing their values have been absorbed into the one term $e_{h,n+1}$. If B is a function space and each function $\hat{y}_{hn}$ is represented by a finite table of values and an interpolation formula, then the interpolation error is also included in e_{hn} .

In the terminology of [4] we are studying the numerical method $\bar{F} = (F, \{F_h\}, \{E_h\}, H, X, Y)$ where X is the data space whose elements are pairs (f, a) , $f: [0, 1] \times B \rightarrow B$, $a \in B$, which yield solutions to (1.1); Y is as already defined; $E_h = B^{N+1}$; $F_h: X \times E_h \rightarrow Y$ is the function defined by (1.4) and (1.5) and an appropriate interpolation formula, where its arguments are $(f, a) \in X$ and $(e_{h0}, e_{h1}, \dots, e_{hN}) \in E_h$; $H = \{h: 1/h = N \text{ is an integer}\}$; and $\phi_h: E_h \rightarrow R$, the critically important function that defines the topology on E_h , is defined by

$$\phi_h(e_{h0}, e_{h1}, \dots, e_{hN}) = \max_{0 \leq k \leq N} \left\| \sum_{i=0}^k e_{hi} \right\|.$$

(This particular ϕ function was first used by Spijker [12].)

Examination of the way the parameters in an R–K method are derived from Taylor series in two variables (see e.g. Butcher [2], Gear [8], Henrici [10] or Ralston [12]) suggests that the actual order of convergence should be sensitive

to the properties of f as well as to the properties of the solution y . (The derivation of a method designed to yield global order of convergence h^n usually requires the assumption that all mixed partial derivatives of f of total order $n = 1$ exist; however, for some exceptions see [6]. This is in contrast to linear multistep methods where, apart from the requirement of Lipschitz continuity in its second argument, the properties of f play no part in determining the order of convergence.) In this paper we examine the extent of this sensitivity. We show that it is less than one would expect from the customary derivations of R-K methods.

Consider as an example the ODE in one real variable

$$(1.6) \quad y'(x) = y + |y - e^x| \sin(r(x)y), \quad y(0) = 1,$$

and its solution $y(x) = \exp(x)$, where $r: [0, 1] \rightarrow \mathbb{R}$ is a bounded function which is nowhere continuous. Observe that the right-hand side is uniformly Lipschitz continuous in y (within a neighborhood of the solution), but it has no partial derivatives with respect to either variable at any point on the solution trajectory. We prove that for any R-K method with design order of convergence at least two, the numerical solution of (1.6) converges to the exact solution with order h^2 ; that we can find R-K methods of design order h^3 which achieve this order on (1.6); and that it is possible to design R-K methods that will achieve any desired order of convergence on (1.6), but only at the expense of using more function evaluations than conventional R-K methods. We also include some numerical experiments to show that our theorems accurately describe the actual order of convergence.

2. Convergence results. We shall use the following basic results in our analysis. (Note: throughout this paper when we refer to Riemann integrable we mean properly Riemann integrable; hence we are implying that the integrand is bounded.)

LEMMA 2.1. *Let $y: [0, 1] \rightarrow B$ be Riemann integrable and let $L: B \rightarrow \mathbb{R}$ be any linear functional with $\|L\| = 1$, then $(Ly): [0, 1] \rightarrow \mathbb{R}$ is uniformly Riemann integrable over all such L , i.e., given $\varepsilon > 0$ there exists a $\delta > 0$, independent of L , such that*

$$(2.1) \quad \left| \sum_{\Pi} (Ly)(r_i) \Delta_i - \sum_{\Pi'} (Ly)(r'_i) \Delta'_i \right| < \varepsilon$$

for any partitions Π and Π' of $[0, 1]$ which have $\max \Delta_i$ and $\max \Delta'_i$ both less than δ .

Proof. The proof follows from the observation that

$$\left| \sum_{\Pi} (Ly)(r_i) \Delta_i - \sum_{\Pi'} (Ly)(r'_i) \Delta'_i \right| \leq \left\| \sum_{\Pi} y(r_i) \Delta_i - \sum_{\Pi'} y(r'_i) \Delta'_i \right\|$$

and the fact that the second expression can be made arbitrarily small by choosing a sufficiently small δ since y is Riemann integrable (see Graves [9]).

LEMMA 2.2. *For each $b \in B$ there exists at least one linear functional $L: B \rightarrow \mathbb{R}$ such that $\|L\| = 1$ and $Lb = \|b\|$.*

Proof. This is a well-known corollary of the Hahn-Banach theorem.

We now proceed to our convergence analysis for R-K methods. The first lemma is a standard result in the analysis of implicit R-K methods.

LEMMA 2.3. Let $f: [0, 1] \times B \rightarrow B$ be Lipschitz continuous in its second variable with Lipschitz constant K . Set

$$M = \max \left\{ \sum_{j=0}^m |\beta_{ij}| : 0 \leq i \leq m \right\};$$

then for $i = 0, \dots, m$, $p = 0, \dots, N - 1$ and $h < 1/(MK)$,

$$(2.2) \quad \|k_{ip}(u) - k_{ip}(v)\| \leq \gamma \|u - v\|,$$

where $\gamma = K/(1 - hMK)$.

Proof. Using the Lipschitz continuity of f we obtain

$$\begin{aligned} \|k_{ip}(u) - k_{ip}(v)\| &= \left\| f \left(t_{h,p+\alpha_i}, u + h \sum_{j=0}^m \beta_{ij} k_{jp}(u) \right) \right. \\ &\quad \left. - f \left(t_{h,p+\alpha_i}, v + h \sum_{j=0}^m \beta_{ij} k_{jp}(v) \right) \right\| \\ &\leq K \left(\|u - v\| + h \sum_{j=0}^m |\beta_{ij}| \|k_{jp}(u) - k_{jp}(v)\| \right). \end{aligned}$$

Set $z = \max \{ \|k_{ip}(u) - k_{ip}(v)\| : 0 \leq i \leq m \}$. Then

$$z \leq K(\|u - v\| + hMz);$$

hence

$$z \leq \gamma \|u - v\|.$$

If we did not need a special form of stability bound (i.e., (2.3)), the proof of the next theorem would be a corollary of the well-known result that a one-step method with Lipschitz continuous increment function is stable and Lemma 2.3.

THEOREM 2.4 (Order-Stability). Suppose the hypothesis of Lemma 2.3 holds, and let $\hat{v}_{hn}$ be the solution to (1.5) when all the e_{hn} are zero; then

$$(2.3) \quad \|\hat{y}_{hn} - \hat{v}_{hn}\| \leq \bar{M} e^{\bar{M}} \max_{0 \leq j \leq N} \left\| \sum_{i=0}^j e_{hi} \right\|, \quad n = 0, \dots, N,$$

where $\bar{M} = \max(1, M^*)$ and $M^* = \gamma \sum_{i=0}^m |w_i|$.

Proof. By definition,

$$\begin{aligned} \hat{y}_{h,j+1} - \hat{v}_{h,j+1} &= \hat{y}_{hj} - \hat{v}_{hj} + h \sum_{i=0}^m w_i (k_{ij}(\hat{y}_{hj}) - k_{ij}(\hat{v}_{hj})) + e_{h,j+1}, \\ j &= 0, \dots, N - 1, \end{aligned}$$

and

$$\hat{y}_{h0} - \hat{v}_{h0} = e_{h0}.$$

Summing over j from zero to n gives

$$\hat{y}_{h,n+1} - \hat{v}_{h,n+1} = h \sum_{j=0}^n \sum_{i=0}^m w_i (k_{ij}(\hat{y}_{hj}) - k_{ij}(\hat{v}_{hj})) + \sum_{j=0}^{n+1} e_{hj}.$$

Taking norms and using Lemma 2.3 yields

$$\begin{aligned} \|\hat{y}_{h,n+1} - \hat{v}_{h,n+1}\| &\leq h\gamma \sum_{j=0}^n \sum_{i=0}^m |w_i| \|\hat{y}_{hj} - \hat{v}_{hj}\| + \left\| \sum_{j=0}^{n+1} e_{hj} \right\| \\ &\leq hM^* \sum_{j=0}^n \|\hat{y}_{hj} - \hat{v}_{hj}\| + \left\| \sum_{j=0}^{n+1} e_{hj} \right\|, \quad n = -1, 0, \dots, N-1. \end{aligned}$$

It now follows easily by induction on n that

$$\|\hat{y}_{hn} - \hat{v}_{hn}\| \leq \bar{M}(1 + h\bar{M})^n \max_{0 \leq j \leq n} \left\| \sum_{i=0}^j e_{hi} \right\|,$$

from which (2.3) immediately follows.

As γ is a function of h , then so is $\bar{M}$, but it is clear that for all $h \leq h_0 < 1/(MK)$, $\bar{M}$ can be replaced by a constant. Thus, in the terminology of [4], Theorem 2.4 states that the method is order-stable.

Up to now we have made no assumptions on the relationships between w_i , α_i and β_{ij} . We now assume that we are dealing with an R-K method of design order at least two. We then have the following standard relationships (see, e.g., Butcher [2] or Gear [8]):

$$(2.4a) \quad \sum_{j=0}^m \beta_{ij} = \alpha_i, \quad 0 \leq i \leq m,$$

$$(2.4b) \quad \sum_{i=0}^m w_i = 1,$$

$$(2.4c) \quad \sum_{i=0}^m w_i \alpha_i = 1/2.$$

DEFINITION 2.5. We say $y:[0, 1] \rightarrow B$ is in class $S_p^R[0, 1]$ ($p \geq 1$) if y is $p-1$ times differentiable on $[0, 1]$ and there exists a function we will denote by $y^{(p)}:[0, 1] \rightarrow B$ such that $y^{(p-1)}$ is the Riemann integral of $y^{(p)}$. We say $y \in S_p^B[0, 1]$ if $y \in S_p^R[0, 1]$ and $y^{(p)}$ is of bounded variation, i.e., if there exists a real constant V such that for any finite partition $0 \leq t_1 < \dots < t_r \leq 1$ we have

$$\sum_{j=1}^r \|y^{(p)}(t_j) - y^{(p)}(t_{j-1})\| \leq V.$$

Examples. (i) Let $B = R$; then each function

$$y_n(x) = (x - 1/2)^n |x - 1/2|, \quad n = 0, 1, 2, \dots,$$

is in $S_{n+1}^B[0, 1]$, but is only n times differentiable on $[0, 1]$.

(ii) Graves [9] proved that if g is the indefinite Riemann integral of f , then f is the derivative of g at every point of continuity of f , but that, in a Banach space, a nowhere continuous function may be Riemann integrable. Hence, a function may be a Riemann integral and yet be nowhere differentiable. Let B be the space of real-valued bounded functions on $[0, 1]$ with supremum norm. Define

$f:[0, 1] \rightarrow B$ by

$$f(t)(x) = \begin{cases} 0 & \text{if } x \leq t, \\ 1 & \text{if } x > t. \end{cases}$$

This is everywhere discontinuous, as $\|f(t) - f(t')\| = 1$ whenever $t \neq t'$. Its Riemann integral

$$g(t) = \int_0^t f(t') dt'$$

is the function

$$g(t)(x) = \begin{cases} x & \text{if } x \leq t, \\ t & \text{if } x > t, \end{cases}$$

Checking to see whether f is the derivative of g , we evaluate

$$\left(\frac{g(t + \Delta t) - g(t)}{\Delta t} - f(t) \right)(x) = \begin{cases} \frac{x - (t + \Delta t)}{\Delta t} & \text{if } t < x \leq t + \Delta t, \\ 0 & \text{otherwise,} \end{cases}$$

and find that the norm of this difference is 1 for all $\Delta t \neq 0$. Thus f is nowhere the derivative of g ; in fact g is nowhere differentiable. Nevertheless, $g \in S_1^R[0, 1]$ (but $g \notin S_1^B[0, 1]$) and, for our purposes, we denote f by g' .

LEMMA 2.6. Suppose $f:[0, 1] \times B \rightarrow B$ is Lipschitz continuous in its second variable, $y \in S_2^R[0, 1]$, and (2.4a) holds; then for $h \leq h_0 < 1/(MK)$ there exists a positive constant c , independent of h, n and i , such that

$$(2.5) \quad \|k_{in}(y_{hn}) - y'_{h,n+\alpha_i}\| \leq ch^2, \quad i = 0, \dots, m,$$

where y'_{hs} denotes $y'(sh)$.

Proof. Let K be the Lipschitz constant of f . By definition

$$\begin{aligned} & \|k_{in}(y_{hn}) - y'_{h,n+\alpha_i}\| \\ &= \left\| f\left(t_{h,n+\alpha_i}, y_{hn} + h \sum_{j=0}^m \beta_{ij} k_{jn}(y_{hn})\right) - f(t_{h,n+\alpha_i}, y_{h,n+\alpha_i}) \right\| \\ &\leq K \left(\left\| y_{h,n+\alpha_i} - y_{hn} - h \sum_{j=0}^m \beta_{ij} y'_{h,n+\alpha_j} \right\| + h \sum_{j=0}^m |\beta_{ij}| \|k_{jn}(y_{hn}) - y'_{h,n+\alpha_j}\| \right). \end{aligned}$$

Now let L be an arbitrary linear function on B with $\|L\| = 1$. Calculate using (2.4a) and integration by parts, using Lemma 2.1, that

$$\begin{aligned} & \left| L\left(y_{h,n+\alpha_i} - y_{hn} - h \sum_{j=0}^m \beta_{ij} y'_{h,n+\alpha_j}\right) \right| \\ &= \left| \int_{t_{hn}}^{t_{h,n+\alpha_i}} (t_{h,n+\alpha_i} - t) L y''(t) dt - h \sum_{j=0}^m \beta_{ij} \int_{t_{hn}}^{t_{h,n+\alpha_j}} L y''(t) dt \right| \\ &\leq \tilde{c} h^2, \end{aligned}$$

where $\tilde{c} = (\frac{1}{2} + M)\|y''\|$. Thus, by Lemma 2.2,

$$\|k_{in}(y_{hn}) - y'_{h,n+\alpha_i}\| \leq \tilde{c}Kh^2 + hKM \max_{0 \leq j \leq m} \|k_{jn}(y_{hn}) - y'_{h,n+\alpha_j}\|.$$

Then, as in Lemma 2.3, we can conclude

$$\|k_{in}(y_{hn}) - y'_{h,n+\alpha_i}\| \leq \tilde{c}\gamma h^2, \quad i = 0, \dots, m, \quad n = 0, \dots, N-1,$$

and the lemma follows.

THEOREM 2.7 (Order of consistency). *Suppose the hypothesis of Lemma 2.6 holds and (2.4b) and (2.4c) also hold; then*

$$(2.6) \quad \max_{1 \leq n \leq N} \left\| \sum_{i=1}^n \tau_{hi} \right\| = o(h).$$

If, in addition, $y \in S_2^B[0, 1]$, then

$$(2.7) \quad \max_{1 \leq n \leq N} \left\| \sum_{i=1}^n \tau_{hi} \right\| = O(h^2).$$

Proof. By definition,

$$\begin{aligned} \tau_{h,i+1} &= y_{h,i+1} - y_{hi} - h \sum_{j=0}^m w_j k_{ji}(y_{hi}) \\ &= y_{h,i+1} - y_{hi} - h \sum_{j=0}^m w_j y'_{h,i+\alpha_j} + h \sum_{j=0}^m w_j (y'_{h,i+\alpha_j} - k_{ji}(y_{hi})). \end{aligned}$$

Let L be an arbitrary linear functional on B with unit norm. Then observing that summation over n in the final term cancels out one power of h , we have

$$\left| L \sum_{i=0}^n \tau_{h,i+1} \right| \leq \tilde{\tau}_{hn} + c^* h^2, \quad n = 0, \dots, N-1,$$

where

$$\begin{aligned} \tilde{\tau}_{hn} &= \left| L \sum_{i=0}^n \left(y_{h,i+1} - y_{hi} - h \sum_{j=0}^m w_j y'_{h,i+\alpha_j} \right) \right|, \\ c^* &= c \sum_{j=0}^m |w_j| \end{aligned}$$

and c is as in Lemma 2.6. Then using (2.4b) and integration by parts, we have

$$\tilde{\tau}_{hn} = \left| \sum_{i=0}^n \left(\int_{t_{hi}}^{t_{h,i+1}} (t_{h,i+1} - t) Ly''(t) dt - h \sum_{j=0}^m w_j \int_{t_{hi}}^{t_{h,i+\alpha_j}} Ly''(t) dt \right) \right|.$$

From (2.4c) we can conclude

$$\begin{aligned} \tilde{\tau}_{hn} &= \left| \sum_{i=0}^n \left(\int_{t_{hi}}^{t_{h,i+1}} (t_{h,i+1} - t) (Ly''(t) - Ly''(t_{hi})) dt \right. \right. \\ &\quad \left. \left. - h \sum_{j=0}^m w_j \int_{t_{hi}}^{t_{h,i+\alpha_j}} (Ly''(t) - Ly''(t_{hi})) dt \right) \right|, \end{aligned}$$

i.e.,

$$(2.8) \quad \tilde{\tau}_{hn} \leq \left(\frac{1}{2} + \sum_{j=0}^m |w_j| \right) h^2 \sum_{i=0}^{N-1} (M_{hi} - m_{hi}),$$

where

$$M_{hi} = \sup \{Ly''(t) : t \in [t_{hi}, t_{h,i+1}]\},$$

$$m_{hi} = \inf \{Ly''(t) : t \in [t_{hi}, t_{h,i+1}]\}.$$

Let $\varepsilon > 0$ be given. Since $y \in S_2^R[0, 1]$, Lemma 2.1 applied to Ly'' yields, for h sufficiently small (independent of L),

$$\tilde{\tau}_{hn} \leq \left(\frac{1}{2} + \sum_{j=0}^m |w_j| \right) h\varepsilon.$$

Now let L be as in Lemma 2.2 to yield

$$\left\| \sum_{i=0}^n \tau_{h,i+1} \right\| \leq \left[\left(\frac{1}{2} + \sum_{j=0}^m |w_j| \right) \varepsilon + c^* h \right] h,$$

from which (2.6) follows. If, in addition, $y \in S_2^B[0, 1]$, then the sum in (2.8) is bounded by the bound V on the variation of y'' on $[0, 1]$, independent of L , so that

$$\left\| \sum_{i=0}^n \tau_{h,i+1} \right\| \leq \left[\left(\frac{1}{2} + \sum_{j=0}^m |w_j| \right) V + c^* \right] h^2,$$

from which (2.7) follows.

THEOREM 2.8. *Suppose f is Lipschitz continuous in its second variable, conditions (2.4a), (2.4b) and (2.4c) hold, and $y \in S_2^R[0, 1]$; then the R-K method converges to y with order $o(h)$. Furthermore, if $y \in S_2^B[0, 1]$, then the convergence is order $O(h^2)$.*

Proof. This theorem is an immediate consequence of Theorems 2.4 and 2.7 and the general theory of [4].

In [5] we remarked on the effect of using the ϕ -function used here as compared with the function

$$\phi_h^*(e_0, \dots, e_N) = \sum_{i=0}^N \|e_i\|$$

which we used in the applications in [4]. In analogy with the effect of using ϕ_h^* in the analysis of Euler's method, we conjecture that if ϕ_h^* were substituted for ϕ_h in this paper, Theorem 2.8 could be proved only if the condition $y \in S_2^R[0, 1]$ were replaced by the more stringent requirement that y' exists and is continuously differentiable almost everywhere. However, it is only in infinite-dimensional Banach spaces that there exist Riemann integrals that are not differentiable almost everywhere.

3. Obtaining higher orders of convergence. We have just shown that all R-K methods designed to yield global order two or higher (i.e., satisfying (2.4)) will realize an order h^2 convergence if the exact solution is sufficiently well-behaved (viz., is in S_2^B) and the right-hand side f is Lipschitz continuous in y , no

matter how badly behaved f otherwise is. The numerical results of the next section show that higher orders of convergence cannot, in general, be obtained purely by further restricting the continuity class of y . However, as proved by the next theorem, for each positive integer q there does exist a restricted class of R-K methods which realizes an actual order of convergence $o(h^q)$ or $O(h^{q+1})$, according as y is in S_{q+1}^R or S_{q+1}^B , for every f that is merely Lipschitz continuous in y . A method in such a class may, if designed to do so, yield an even higher order of convergence when f is of a higher continuity class.

DEFINITION 3.1. Let $T_0 = \{0, 1, \dots, m\}$. For $q = 1, 2, 3, \dots$, let T_q be the set containing every $i \in T_{q-1}$ which satisfies

$$\sum_{j=0}^m \beta_{ij} \alpha_j^{q-1} = \frac{\alpha_i^q}{q}$$

and

$$\beta_{ij} = 0 \quad \text{for every } j \notin T_{q-1}.$$

LEMMA 3.2. Let $f: [0, 1] \times B \rightarrow B$ be Lipschitz continuous in its second variable with Lipschitz constant K and suppose $T_1 = T_0$ (i.e., (2.4a) holds). If $y \in S_{q+1}^R[0, 1]$; then

$$\|k_{in}(y_{hn}) - y'_{h,n+\alpha_i}\| = O(h^{q+1})$$

for every $i \in T_q$.

Proof. We proceed by induction on q . The case $q = 1$ is just Lemma 2.6. Suppose $i \in T_p$ and the induction hypothesis holds for $q = p - 1$. Then as in Lemma 2.6,

$$\begin{aligned} & \|k_{in}(y_{hn}) - y'_{h,n+\alpha_i}\| \\ & \leq K \left(\left\| y_{h,n+\alpha_i} - y_{hn} - h \sum_{j \in T_{p-1}} \beta_{ij} y'_{h,n+\alpha_j} \right\| + hM \max_{j \in T_{p-1}} \|k_{jn}(y_{hn}) - y'_{h,n+\alpha_j}\| \right). \end{aligned}$$

Proceeding as in Lemma 2.6, using Taylor series with integral remainder for the first term and the induction hypothesis for the second

$$\begin{aligned} & \|k_{in}(y_{hn}) - y'_{h,n+\alpha_i}\| \\ & \leq K \left[\frac{1}{p!} \int_{t_{hn}}^{t_{h,n+\alpha_i}} (t_{h,n+\alpha_i} - t)^p L y^{(p+1)}(t) dt \right. \\ & \quad \left. - \frac{h}{(p-1)!} \sum_{j \in T_{p-1}} \beta_{ij} \int_{t_{hn}}^{t_{h,n+\alpha_j}} (t_{h,n+\alpha_j} - t)^{p-1} L y^{(p+1)}(t) dt \right] + O(h^{p+1}) \\ & \leq K \left(\frac{1}{p+1} + M \right) \|y^{(p+1)}\| h^{p+1}/p! + O(h^{p+1}) = O(h^{p+1}). \end{aligned}$$

THEOREM 3.3 (Order of consistency). Suppose the hypothesis of Lemma 3.2 holds, and suppose for some $q \geq 1$,

$$(3.1a) \quad w_j = 0 \quad \text{for all } j \notin T_q,$$

$$(3.1b) \quad \sum_{j \in T_q} \alpha_j^p w_j = \frac{1}{p+1} \quad \text{for } p = 0, 1, \dots, q.$$

Then

$$\max_{1 \leq n \leq N} \left\| \sum_{i=1}^n \tau_{hi} \right\| = \begin{cases} o(h^q) & \text{if } y \in S_{q+1}^R[0, 1], \\ O(h^{q+1}) & \text{if } y \in S_{q+1}^B[0, 1]. \end{cases}$$

Proof. In a manner analogous to Theorem 2.7,

$$\left| L \sum_{i=0}^n \tau_{h,i+1} \right| \leq \tau_{hn}^{(1)} + \tau_{hn}^{(2)},$$

where

$$\begin{aligned} \tau_{hn}^{(1)} &= \left| L \sum_{i=0}^n \left(y_{h,i+1} - y_{hi} - h \sum_{j \in T_q} w_j y'_{h,i+\alpha_j} \right) \right|, \\ \tau_{hn}^{(2)} &= h \left| L \sum_{i=0}^n \sum_{j \in T_q} w_j (y'_{h,i+\alpha_j} - k_{ji}(y_{hi})) \right| \\ &= O(h^{q+1}) \quad (\text{by Lemma 3.2}). \end{aligned}$$

Using the Taylor series with integral remainder on Ly and (3.1b), we obtain

$$\begin{aligned} \tau_{hn}^{(1)} &= \left| \sum_{i=0}^n \left(\frac{1}{q!} \int_{t_{hi}}^{t_{h,i+1}} (t_{h,i+1} - t)^q Ly^{(q+1)}(t) dt \right. \right. \\ &\quad \left. \left. - \frac{h}{(q-1)!} \sum_{j \in T_q} w_j \int_{t_{hi}}^{t_{h,i+\alpha_j}} (t_{h,i+\alpha_j} - t)^{q-1} Ly^{(q+1)}(t) dt \right) \right|. \end{aligned}$$

Using (3.1b) again with $p = q$ yields

$$\begin{aligned} \tau_{hn}^{(1)} &= \left| \sum_{i=0}^n \left(\frac{1}{q!} \int_{t_{hi}}^{t_{h,i+1}} (t_{h,i+1} - t)^q (Ly^{(q+1)}(t) - Ly^{(q+1)}(t_{hi})) dt \right. \right. \\ &\quad \left. \left. - \frac{h}{(q-1)!} \sum_{j \in T_q} w_j \int_{t_{hi}}^{t_{h,i+\alpha_j}} (t_{h,i+\alpha_j} - t)^{q-1} (Ly^{(q+1)}(t) - Ly^{(q+1)}(t_{hi})) dt \right) \right|. \end{aligned}$$

Hence

$$\tau_{hn}^{(1)} \leq \left(\frac{1}{q+1} + \sum_{j \in T_q} |w_j| \right) (h^{q+1}/q!) \sum_{i=0}^n (M_{hi} - m_{hi}),$$

where

$$\begin{aligned} M_{hi} &= \sup \{ Ly^{(q+1)}(t) : t \in [t_{hi}, t_{h,i+1}] \}, \\ m_{ih} &= \inf \{ Ly^{(q+1)}(t) : t \in [t_{hi}, t_{h,i+1}] \}. \end{aligned}$$

The rest of the proof is just like that of Theorem 2.7.

THEOREM 3.4. Assume the hypotheses of Theorem 3.3 hold. Then the R - K method converges like $o(h^q)$ if $y \in S_{q+1}^R[0, 1]$ and like $O(h^{q+1})$ if $y \in S_{q+1}^B[0, 1]$.

Proof. This theorem is an immediate consequence of Theorems 2.4 and 3.3 and the general theory of [4].

Theorem 3.3 has some interesting consequences for explicit R-K methods in which, typically,

$$\begin{aligned}\beta_{ij} &= 0 \quad \text{whenever } j \geq i, \\ \alpha_0 &= 0, \\ T_0 = T_1 &= \{0, 1, 2, 3, \dots\}, \\ T_2 &= \{0, 2, 3, \dots\}, \quad T_3 = \{0, 3, \dots\}, \quad \text{etc.}\end{aligned}$$

Condition (3.1a) then tells us that we will achieve convergence of order $o(h^q)$ or $O(h^{q+1})$, given the corresponding condition on y , if we use a method in which $w_i = 0$ for $i = 1, 2, \dots, q - 1$. Examples of such methods for $q = 2$ and $q = 3$ are found in the next section. The Runge-Kutta-Merson method (see Gear [8, p. 85]) is an example of a method with $w_1 = w_2 = 0$. The numerical experiments in the next section (admittedly, limited to R^1 , an especially simple Banach space) suggest that this condition on the w_i is necessary as well as sufficient to guarantee the order of convergence when f has no partial derivatives. The interesting question of how these results are affected when f has a limited number of partial derivatives is one we have made no attempt to investigate. (However, see [6] for some results in this area.) It seems possible that a continued study of this problem might lead to the development of new criteria for choosing the parameters of R-K methods based on the minimization of their sensitivity to various types of singularities in the differential equation.

The question of how the convergence of a R-K method is affected by singular behavior of the solution y is the subject of the following theorem, whose proof is omitted as it is so similar to the proofs of Theorems 2.7 and 3.3.

THEOREM 3.5. *Let $f: [0, 1] \times B \rightarrow B$ be Lipschitz continuous in its second variable; then any R-K method which satisfies (2.4a) and (2.4b) converges like $o(1)$ if $y \in S_1^R[0, 1]$, i.e., if $f(t, y(t))$ is Riemann integrable, and like $O(h)$ if $y \in S_1^B[0, 1]$, i.e., if $f(t, y(t))$ is of bounded variation.*

This theorem applies, in particular, to Euler's method and hence extends the result obtained in [3] for the case of a single real-valued ODE to Euler's method for ODE's in Banach spaces.

4. Numerical results. We report here a sample of results obtained by applying several explicit R-K methods to the ODE (1.6). These computations were performed on a CDC 6400, using double-precision where necessary. The function $r(x)$ was evaluated as the sixth term of the sequence of pseudo-random numbers generated by the FORTRAN RANF function using $X + 1.0$ as the "seed". We use the tabular representation of R-K methods used by Hull et al. [11], viz.,

$$\begin{array}{c|cccc} \alpha_0 & \beta_{00} & \beta_{01} & \cdots & \beta_{0m} \\ \alpha_1 & \beta_{10} & \beta_{11} & \cdots & \beta_{1m} \\ \vdots & & & & \\ \vdots & & & & \\ \alpha_m & \beta_{m0} & \beta_{m1} & \cdots & \beta_{mm} \\ \hline & w_0 & w_1 & & w_m \end{array}$$

where, as only explicit methods were used, each β -matrix is lower triangular with zero diagonal.

The general two-stage second order R-K method is the following.

Method A.

$$\begin{array}{c|cc} 0 & & \\ a & & a \\ \hline & 1 - 1/(2a) & 1/(2a) \end{array}$$

We tested the cases $a = 1/2$ and 1, and obtained $O(h^2)$ convergence, as predicted by Theorem 2.8, in both cases. Method A of Table 4.1 is the case $a = 1$.

TABLE 4.1
The error at $x = 1$ in the computed solution of (1.6) using various explicit R-K methods described in the text

$-\log_2 h$	$-\log_2 \text{error} $					
	A	B	C	D	E	F
0	5.0	5.2	2.7	6.9	2.6	12.2
1	5.6	9.2	5.1	10.6	6.1	14.7
2	9.2	12.0	6.3	11.9	6.5	17.6
3	11.6	14.8	8.1	14.8	8.7	21.6
4	14.6	17.9	10.0	17.7	10.5	25.4
5	15.7	20.8	11.9	20.6	12.6	29.8
6	18.8	23.5	13.9	23.6	14.6	33.7
7	20.5	26.9	15.9	26.7	16.7	37.7
8	23.4	29.4	18.0	29.8	18.7	44.8
9	24.8	32.5	20.0	32.9	20.6	45.5

The general three-stage third order R-K method with $w_1 = 0$ is the following.

Method B.

$$\begin{array}{c|ccc} 0 & & & \\ a & & & a \\ 2/3 & 2/3 - 2/(9a) & 2/(9a) & \\ \hline & 1/4 & 0 & 3/4 \end{array}$$

This satisfies the hypothesis of Theorem 3.4 with $q = 2$, $T_2 = \{2\}$ for all a , hence should yield $O(h^3)$ convergence. This was experimentally verified for $a = 1/3$ and $2/3$. The case $a = 1/3$ is reported in Table 4 as Method B.

To demonstrate the effect of a nonzero w_1 , we tested the classical third order method below.

Method C.

$$\begin{array}{c|ccc} 0 & & & \\ 1/2 & 1/2 & & \\ 1 & -1 & 2 & \\ \hline & 1/6 & 2/3 & 1/6 \end{array}$$

This yielded only $O(h^2)$ convergence.

There is no four-stage fourth order explicit method with $w_1 = w_2 = 0$, hence no method of this class that will yield $O(h^4)$ convergence on (1.6). The general method of this class with $w_1 = 0$ is the following.

Method D.

0				
a	a			
$1/2$	$1/2 - 1/(8a)$	$1/(8a)$		
1	$1/(2a) - 1$	$-1/(2a)$	2	
	$1/6$	0	$2/3$	$1/6$

This satisfies the hypothesis of Theorem 3.4 with $q = 2$, $T_2 = \{0, 2, 3\}$; hence we expect $O(h^3)$ convergence for all a . This was verified for $a = 1/4$ and $1/2$. The case $a = 1/2$ is reported in Table 4.1 as Method D.

For comparison, we report also the classical fourth order method as follows.

Method E.

0				
$1/2$	$1/2$			
$1/2$	0	$1/2$		
1	0	0	1	
	$1/6$	$1/3$	$1/3$	$1/6$

This yielded $O(h^2)$ convergence.

The Runge-Kutta-Merson method (Gear [8, p. 85]) is the five-step fourth order method, which follows.

Method F.

0					
$1/3$	$1/3$				
$1/3$	$1/6$	$1/6$			
$1/2$	$1/8$	0	$3/8$		
1	$1/2$	0	$-3/2$	2	
	$1/6$	0	0	$2/3$	$1/6$

This satisfies the hypothesis of Theorem 3.4 with $q = 3$, $T_2 = \{0, 2, 3, 4\}$, $T_3 = \{0, 3, 4\}$ and hence yields $O(h^4)$ convergence on our test problem, as shown in Table 4.1.

Table 4.1 and 4.2 exhibit a noteworthy phenomenon in the erratic behavior of the error. Although the error is of the order in h predicted by the theory, it is clearly not proportional to any power of h . This means that we could not expect to improve any of the computed results by using h -extrapolation. (A similar result was reported by Feldstein and Goodman [7] in the application of Euler's method

TABLE 4.2

$-\log_2 h$	$p = \log_2 \text{error}(h) - \log_2 \left \text{error} \left(\frac{h}{2} \right) \right $					
	A	B	C	D	E	F
0	0.6	4.0	2.4	3.7	3.5	2.5
1	3.6	2.8	1.2	1.3	0.4	2.9
2	2.4	2.8	1.8	2.9	2.2	4.0
3	3.0	3.1	1.9	2.9	1.8	3.8
4	1.1	2.9	1.9	2.9	2.1	4.4
5	3.1	2.7	2.0	3.0	2.0	3.9
6	1.7	3.4	2.0	3.1	2.1	4.0
7	2.9	2.5	2.1	3.1	2.0	7.1
8	1.4	3.1	2.0	3.1	1.9	0.7

to retarded differential equations with discontinuities in the right-hand side.) Table 4.2 exhibits explicitly the variation in p if we assume that $\text{error}(h) = Ch^p$.

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